THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): A body can have zero displacement but non-zero distance. Reason (R): Distance is the length of the path and displacement is the net change in position. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A body is thrown upward with velocity u. The velocity at half the maximum height is: 1

(a) u / 2 **(b)** u / sqrt(2) **(c)** u / 4 **(d)** u

3. A boat goes 12 km downstream in 1 hour and returns upstream in 2 hours. The average speed for the round trip is: 1

(a) 8 km/h **(b)** 6 km/h **(c)** 12 km/h **(d)** 9 km/h

4. Assertion (A): A body moving in a circle with constant speed is accelerated. Reason (R): The direction of its velocity changes continuously. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A car goes from rest with acceleration a for time t, then retards uniformly with 2a till it stops. The total time is: 1

(a) 1.5 t **(b)** 3 t **(c)** t **(d)** 2 t

6. A man walks to a market at 4 km/h and returns at 6 km/h. The average speed is: 1

(a) 5 km/h **(b)** 5.5 km/h **(c)** 4.8 km/h **(d)** 4.5 km/h

7. Assertion (A): The average velocity of a body in a round trip is zero. Reason (R): The net displacement is zero. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A stone dropped from a height h takes time t. If dropped from 4h, the time taken will be: 1

(a) t **(b)** t/2 **(c)** 2t **(d)** 4t

9. A body covers the first half of its journey at 20 m/s and the second half at 30 m/s. The average speed is: 1

(a) 25 m/s **(b)** 24 m/s **(c)** 26 m/s **(d)** 50 m/s

10. Assertion (A): A ball thrown vertically up has zero acceleration at the highest point. Reason (R): Its velocity is zero at the highest point. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A car travels the first half of the distance at 30 km/h and the next half at 60 km/h. Find the average speed. 2

12. A body is dropped from rest and also another thrown down with 10 m/s from the same height. Compare the time of fall if h = 20 m (g = 10). 2

13. A bus moving at 15 m/s sees a red light 60 m ahead. Find the minimum retardation needed to stop it just before the light. 2

14. A ball is thrown vertically upward and returns to the thrower in 4 s. Find the speed of throw and the maximum height (g = 10). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A particle starts from rest with a = 4 m/s squared for 5 s, then moves uniformly for 10 s and then decelerates at 2 m/s squared until it stops. Find the total distance. 3

16. A ball rolls down an incline from rest and covers 2 m in the 1st second. Find the acceleration and the distance in 4 s. 3

17. A lift moves upward with acceleration 2 m/s squared for 4 s, then uniformly for 6 s and decelerates at 4 m/s squared. Find the height travelled. 3

OR

A car A at 90 km/h overtakes car B moving at 72 km/h. If A is 50 m behind B, find the time taken to be 50 m ahead of B. 3

18. A man jumps from a height of 5 m. Find the time of fall and the speed on landing, and the retardation if he stops in 0.1 s (g = 10). 3

OR

A car travelling at 25 m/s is overtaking another at 15 m/s on a road. A truck coming from the opposite direction at 20 m/s is 400 m ahead. Will the overtaking car pass safely if it needs 6 s? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A train starts from rest at station A and stops at station B 3 km away. It accelerates at 0.5 m/s squared for the first part and retards at 1 m/s squared for the last part with no uniform stretch. Find the maximum speed, the total time and the distances in the two parts. 5

OR

Explain the motion of a ball moving in a vertical circle of radius 1 m with speed 4 m/s, the period, the speed vs velocity, and the average velocity over half a period (pi = 3.14). 5

20. A stone is dropped from a balloon rising upwards with 10 m/s when it is at 75 m. Find the time taken by the stone to reach the ground and its velocity then (g = 10). 5

OR

A police car at rest starts moving with 2 m/s squared acceleration when a speeding car moving at 20 m/s passes it. Find where and when the police car catches it and the speed of the police car then. 5

END OF QUESTION PAPER — Total: 40 Marks